BUSINESS PLAN

Lumi-Project-Overview

Robotic Guide Cane

“If AI can have warmth,
I hope it warms these unseen roads first.”

XIAO SIMA | 2026
Email：szemasiu@gmail.com / Phone：13530090306 / WeChat：nicksima / X ： @simaxiao

The Safety Gap in Mobility Is Growing

People with vision impairment (global)

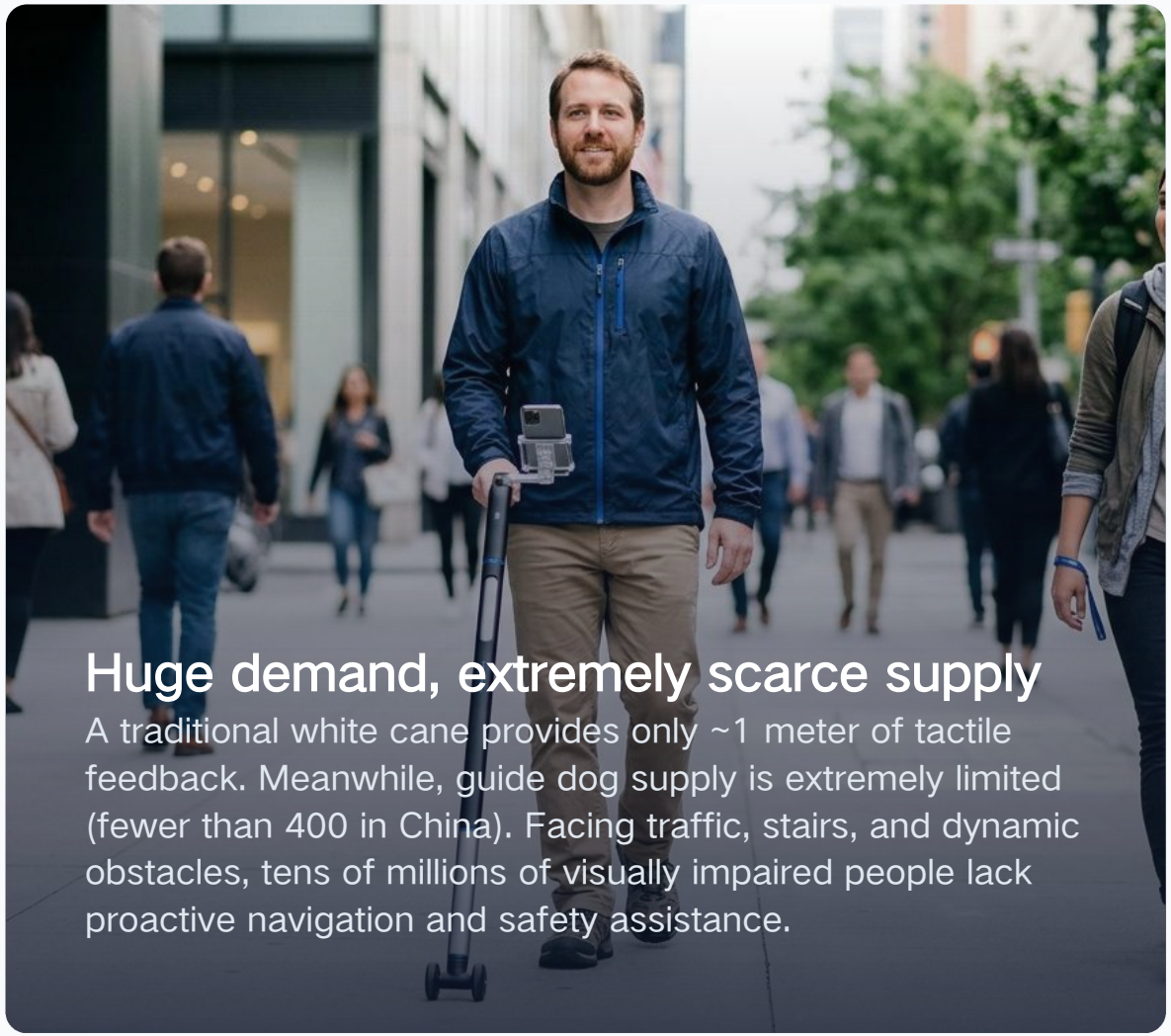
1.1B people, including 295M moderate-to-severe

People with visual impairment in China

17.31M total; ~6 – 8M meet the strict definition of blindness

Active guide dogs in China

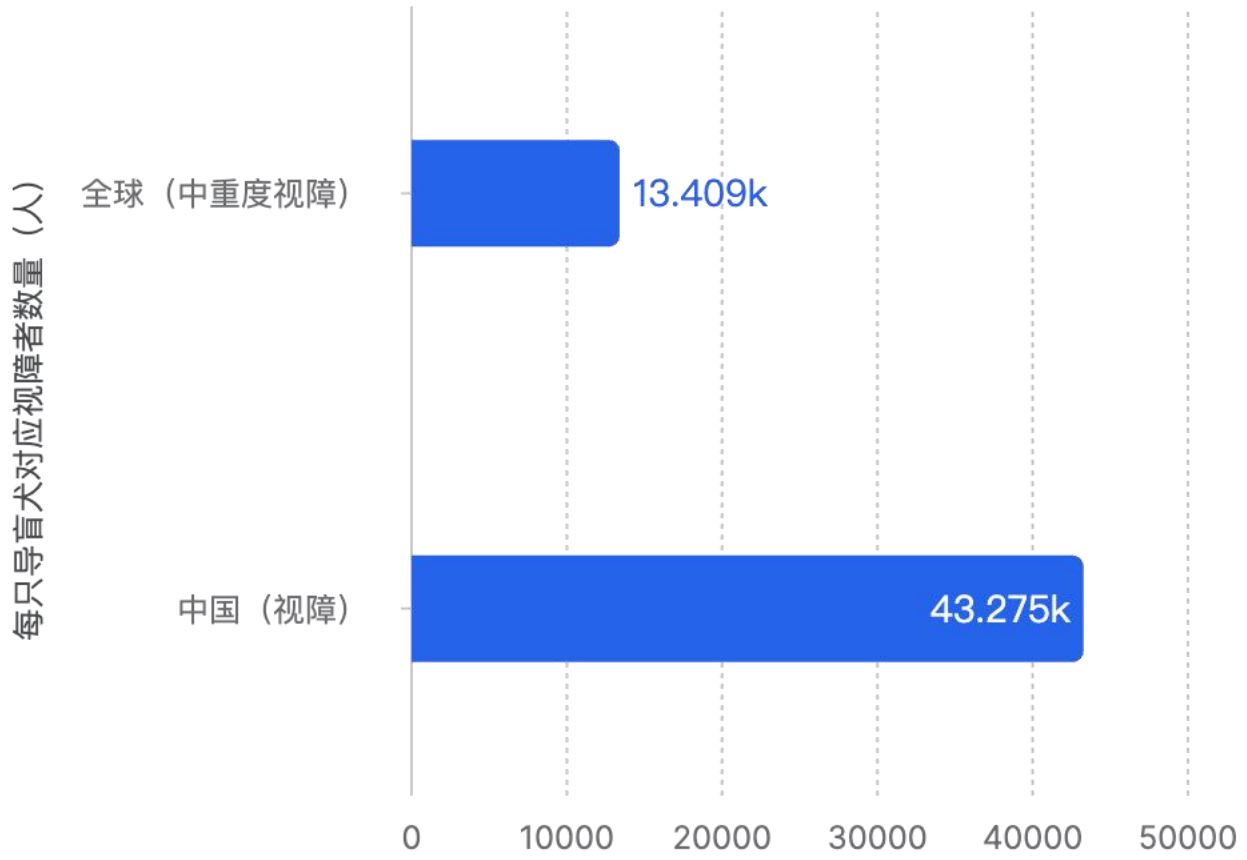
<400 (harder to qualify than becoming a pilot)



Huge demand, extremely scarce supply

A traditional white cane provides only ~1 meter of tactile feedback. Meanwhile, guide dog supply is extremely limited (fewer than 400 in China). Facing traffic, stairs, and dynamic obstacles, tens of millions of visually impaired people lack proactive navigation and safety assistance.

Guide Dog Supply Is Severely Imbalanced



Biological supply has a clear ceiling

Breeding constraints, attrition, and an 18-month professional training cycle make guide dogs extremely expensive (~RMB 200－300K) and impossible to scale through replication.

In China today, roughly one guide dog serves every 43,275 visually impaired people—an enormous gap. Without “technology as public good” and industrial-scale manufacturing, most visually impaired users will never receive high-level mobility safety support.

Core insight:

The market needs a solution that can be standardized and mass-produced like consumer electronics.

Blind Spots & Limitations of Existing Options

White Cane

- **Low cost & high adoption**
  Affordable and widely used
- **Extremely limited sensing range**
  Relies on touch only; ~1 meter ahead
- **No proactive avoidance**
  Cannot handle hanging obstacles, stairs, traffic, and dynamic hazards
- **No route navigation**
  Depends on memory; cannot reach unfamiliar places

Guide Dog

- **Excellent experience & safety**
  Smart avoidance and emotional companionship
- **Extremely expensive**
  RMB 200 – 300K training cost; unaffordable for most users
- **Cannot scale**
  18-month training cycle with high dropout rate
- **High maintenance burden**
  Caring for a large dog is difficult for blind users

Lumi RoadLight

- **Affordable, around RMB 1,000**
  Reuse phone compute; remove expensive hardware
- **Proactive visual avoidance**
  Detects stairs, intersections, crowds, and brakes when needed
- **Industrial-scale manufacturing**
  Standardized assembly; hundreds of units per day
- **Seamless indoor/outdoor navigation**
  GPS + visual SLAM for end-to-end coverage

Phone–Cane Collaboration: Roles & Workflow

The phone “sees & computes”; the cane “moves & stops”

1

App pairing

The user pairs the smartphone App with the robotic cane via Bluetooth/Wi-Fi.

2

Voice destination input

Call out a destination; the phone uses GPS and map APIs to plan the global route.

3

Physical mounting

The phone attaches to the cane via a magnetic mount and can be charged by the cane battery.

4

Real-time sensing & decisions

The phone camera continuously captures the road ahead and detects stairs and obstacles.

5

Chassis motor execution

The phone sends strategies; the cane motors guide the user: forward / turn / emergency stop.

6

Hazard & navigation voice prompts

Real-time voice prompts like “turn right in 3 meters” and “watch the step” complete the closed loop.

Decoupled HW/SW: Affordable hardware + software that keeps improving

Hardware: Standard actuator terminal

- **Chassis powertrain**

BLDC motor + gearbox and dual wheels

- **Energy & control**

2000 – 3000mAh battery with Wi - Fi / Bluetooth modules

- **Minimal structure**

Lightweight aluminum shaft + multi-angle magnetic phone mount

Ultra-low BOM; easy to mass-produce and scale down-market

+

Software: compute & perception brain

- **Visual perception (camera reuse)**

On-device models use the phone camera for obstacle avoidance and scene understanding

- **Global navigation (GPS reuse)**

High-precision map APIs plan outdoor routes

- **Voice interaction (microphone reuse)**

Natural language commands, with real-time warnings and guidance

Evolves with smartphones and frontier models—capabilities keep growing

Core Moat: Tiered Proactive Safety

Reuse smartphone compute and mature AI vision models (e.g., TensorFlow Lite) to handle stairs, traffic, and dynamic hazards.

L1

Long-range prediction: detect static obstacles within 5m

Decision: Auto-decelerate — Motor speed drops (e.g., to 0.3m/s) with early voice alerts.

L2

Mid-range response: dynamic hazards enter the 3m range

Decision: Detour routing — Visual sensing triggers differential wheel control for smooth evasion.

L3

Close-range defense: sudden hazards < 0.5m

Decision: Emergency stop — Motors lock within 0.1s and brakes engage to prevent falls.

Tech Route: Decoupled Layers, Open Ecosystem

C. App & Algorithms: The ecosystem brain

Built by global developers and AI companies using phone compute: visual SLAM, traffic light detection, multimodal navigation, and fast local adaptation.

B. Firmware + API: Standard control plane

Defines standard interfaces (direction / speed / haptics / telemetry). Regardless of algorithm changes, it handshakes with hardware via BLE/Wi - Fi protocols.

A. Hardware layer: Safety-backed actuator

Dual-wheel differential drive, brake locks, IMU and anti-fall structure. Built-in collision protection and link-loss emergency braking ensure no physical harm even when algorithms fail.

Ecosystem Strategy: Open API & Community

Build the hardware infrastructure, let the world build the algorithms

AI and vision models evolve daily. Closed-door development cannot keep pace with the era.
Lumi RoadLight is committed to building an **“extremely low-cost, absolutely safe”** universal chassis. With open APIs and open-source reference code on GitHub, we invite global developers and AI companies to co-create guide Apps. True value naturally brings sustainable business and social respect.

1. Open control interfaces (Robot-as-a-Peripheral)

Developers can control the hardware like a game controller via BLE—standard commands for speed, direction, and haptics.

2. Plugin market & developer ecosystem

Encourage localized optimizations, multi-language packs, and advanced plugins for scenarios like subways and malls.

3. Non-negotiable hardware-level safety

Firmware enforces top-priority safety. If software loses connection or sends dangerous commands, hardware unconditionally locks the brakes—never putting users at risk.

The Core Competitiveness

1

Extreme Accessibility

Making it “affordable”

Traditional robots cost thousands; guide dogs are a luxury at RMB 200－300K. Lumi RoadLight shifts expensive compute and sensing to the smartphone (an existing asset), driving the cane BOM down to the hundreds-RMB level.

2

Industrial Scalability

Free from training cycles

Breeding and an 18-month training cycle create a hard bottleneck for guide dogs. Lumi RoadLight uses standardized mechanical and electronic components—once finalized, it can be mass-produced on assembly lines to meet huge global demand.

3

Continuous Evolution

Data & model flywheel

With App updates, the product never obsoletes. It can seamlessly adopt the latest on-device multimodal models to improve recognition in complex scenes (traffic lights, temporary construction barriers), making the hardware smarter over time.

Go-to-Market Strategy (GTM)

Phase 1: Pilot validation

Partner with blind associations, nonprofits, and ophthalmology departments for targeted early deployments and a closed-loop safety data pipeline.

Phase 2: Channel replication

Enter government/enterprise procurement and accessible city programs (B2B/G). Co-promote as an ecosystem product with phone makers and map service providers.

Phase 3: Scaled breakout

Push to mass consumer markets with “ultra-low hardware buyout + premium software subscription”, reaching broader lower-tier and developing-country markets.

Business Model: From Hardware to Ecosystem Platform

Hardware base sales (the base)

Sell the “smart cane actuator” at an inclusive price close to BOM. Build a massive installed base fast and capture the hardware entry point for the global visually impaired community.

Developer ecosystem & certification

Open APIs attract developers and AI firms to build vision avoidance apps. We provide safety certification and distribution, earning revenue shares from premium plugins.

B2B/G bulk procurement

Stay aligned with Tech for Good. Participate in disability-assistance procurement lists and enterprise CSR projects for bulk delivery with deployment services.

Cloud & premium services

Subscriptions like family cloud monitoring, travel accident insurance linkage, and personalized voice packs build stable recurring revenue (ARR).

Roadmap: From Tech Validation to Full-Scenario Coverage

Demo prototype

Assemble hardware and comms protocols; validate the feasibility of the core stack.

Gen 1: Core productization

Deliver a full loop: visual SLAM + GPS navigation + voice prompts + basic avoidance.

Gen 2: Enhanced avoidance

Add low-cost ultrasonic/ToF sensors and fuse with vision for edge cases like glass and mirrors.

Gen 3: All-weather vision

Adopt on-device multimodal LLMs to improve robustness in low light, rain/snow, and other complex conditions.

Compliance & Governance: Build Safety & Liability into the Product

Safety & legal liability

Device failure or delayed decisions may cause collisions or falls, leading to claims.

Mitigation: Dual-layer safety + risk-sharing

Software enforces a non-crossable 0.5m danger zone; hardware adds an independent electromagnetic brake. Commercially, bundle first-year travel accident insurance to share risk with insurers.

Long-tail misrecognition

Pure vision can fail under strong backlight, rain/snow, or transparent glass walls.

Mitigation: Multimodal redundancy + data flywheel

Gen-2 hardware adds low-cost ToF/ultrasonic ranging. The App includes one-tap reporting; low-confidence scenes trigger voice warnings like "complex environment—please probe manually."

Battery life & device compatibility

Running camera + GPS + models continuously may cause heat and rapid battery drain.

Mitigation: Back-power + model pruning

The cane includes a large battery pack; the mount provides charging contacts. Algorithms use aggressively pruned models (e.g., YOLOv5 Lite), prioritizing mid/low-end Android devices for smooth performance.

Co-creation Needs: Trade Resources for Validation & Scale

Seeking partners to light the road together

Beyond pure financial investment, Lumi RoadLight seeks pioneers across industry, nonprofits, and the developer ecosystem.
We want to put early devices in users' hands, chassis in engineers' hands, and open-source code in developers' hands—bridging the gap from demo to mass production together.

Supply chain & mechanical engineering partners

Need tooling support, dual-wheel chassis engineering optimization, motor prototyping, and durability testing resources under extreme conditions.

Pilot scenarios & nonprofit partners

Looking for organizations and rehab institutions to support the first 100-user pilot and help close the loop on real-world feedback and safety metrics.

Developers & AI ecosystem partners

Seeking on-device compute collaboration with phone makers, deeper map API integration, and SLAM/multimodal developers to contribute plugins and code.

Thanks.

Make AI an accessible public capability — not a new inequality.

https://github.com/simaxiao/Lumi-guide-robot

Contact: XIAO SIMA (Product Manager)
Email：szemasiu@gmail.com / Phone：13530090306 / WeChat：nicksima / X：@simaxiao